香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

ECO6102 / 2026–27 Term 1 (Fall)

Lecture 2: Conditional Expectations

Enjie (Jack) Ma

Lecture 2 · Monday 14 September 2026

Your survey

154 responses — thank you

- **English**: 69 of 154 respondents (44.8%) chose 3/5 for comfort with English lectures.
- **Your interests**: trade, environment, macro, labour, and AI-assisted economic research.
- **Assessment**: problem sets include Python programming; exams test econometric concepts and reasoning, without requiring Python code.

Roadmap

From conditional averages to regression

L1 | L2 | L3 | L4 | L5 | Midterm | L6 | L7 | L8 | Final

Conditional Expectations — Wooldridge (2010), Chapter 2

1. Conditional expectations and models
2. Partial effects and elasticities
3. The error form
4. Properties of conditional expectations and LIE
5. Average partial effects, unobserved regressors and proxies

SECTION 1 · Wooldridge 2.1–2.2.1

Conditional expectations and models

Terminology

Wooldridge 2.1 — one outcome, several explanatory variables

$$y, \qquad \mathbf{x} = (x_1, x_2, \ldots, x_K)$$

- y: explained variable, dependent variable, regressand, or response.
- $\mathbf{x}$: explanatory variables, independent variables, regressors, controls, or covariates.

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- y: explained variable, dependent variable, regressand, or response.
- $\mathbf{x}$: explanatory variables, independent variables, regressors, controls, or covariates.
- Example: $y = \textit{wage}$; $\mathbf{x}$ includes schooling, experience and IQ.

The structural conditional expectation

Wooldridge 2.1 — the relationship we want to learn

How does schooling relate to mean wages when the relevant controls are held fixed?

$$E(y \mid s, \mathbf{c})$$

- s: schooling; $\mathbf{c}$: relevant controls, potentially including ability.

The structural conditional expectation

Wooldridge 2.1 — the relationship we want to learn

How does schooling relate to mean wages when the relevant controls are held fixed?

$$E(y \mid s, \mathbf{c})$$

- s: schooling; $\mathbf{c}$: relevant controls, potentially including ability.
- A causal interpretation needs identification assumptions about the economic setting and controls.
- With suitable data, we can estimate the CE. A parametric functional form makes that task simpler.

Complications

Wooldridge 2.1 — economic data do not always reveal the target

- Measurement error: we observe an imperfect version of a variable.
- Simultaneity: outcomes and explanatory variables may be determined together.
- Selection: the available sample may not represent the population.
- Unobserved controls: ability may affect wages and schooling, but remain unmeasured.

Why conditional expectations?

Which average can the data reveal, and which average do we want?

- **Target**: mean wages at fixed schooling, experience *and ability*: $E(wage \mid educ, exper, ability)$.

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- **Available data**: perhaps we can group workers only by schooling and experience: $E(wage \mid educ, exper)$.

Why conditional expectations?

Which average can the data reveal, and which average do we want?

- **Target**: mean wages at fixed schooling, experience *and ability*: $E(wage \mid educ, exper, ability)$.
- **Available data**: perhaps we can group workers only by schooling and experience: $E(wage \mid educ, exper)$.
- CE notation distinguishes the two. Its properties and additional assumptions help us relate them.

Everything right of the bar is the information set, what we know when we form the average: $E(y \mid \text{information})$.

Conditional expectation

Wooldridge 2.2.1 — a mean for each group of conditioning values

For an outcome Y with $E(|Y|) < \infty$, write

$$E(Y \mid \mathbf{X}) = \mu(\mathbf{X}).$$

- The function μ assigns an average outcome to each set of conditioning values.

Conditional expectation

Wooldridge 2.2.1 — a mean for each group of conditioning values

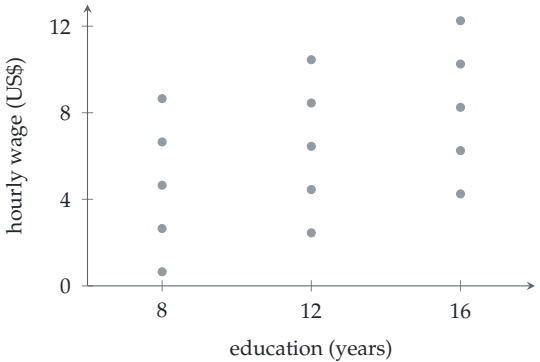
For an outcome Y with $E(|Y|) < \infty$, write

$$E(Y \mid \mathbf{X}) = \mu(\mathbf{X}).$$

- The function μ assigns an average outcome to each set of conditioning values.
- $E(wage \mid educ, exper, IQ)$: average wage among workers with those characteristics.
- Individual wages can differ even when these characteristics are the same.

Conditional expectation: worked example

Illustrative population — average within each schooling group

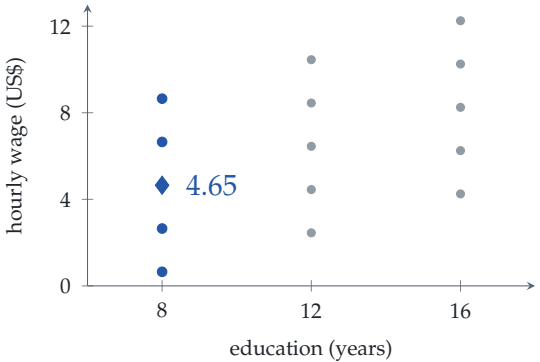


- X: schooling; Y: wage. Five workers at each schooling level.

Illustrative population; Wooldridge 2.2.1.

Conditional expectation: worked example

Illustrative population — average within each schooling group

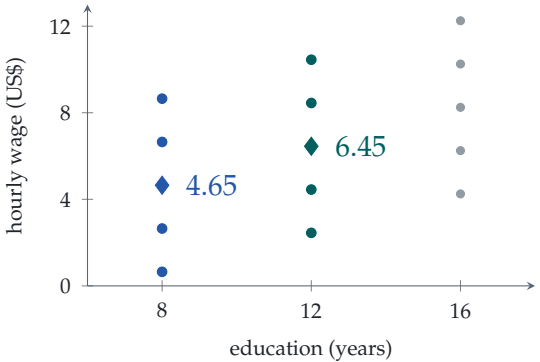


- X: schooling; Y: wage. Five workers at each schooling level.
- $X = 8$: group mean 4.65.

Illustrative population; Wooldridge 2.2.1.

Conditional expectation: worked example

Illustrative population — average within each schooling group

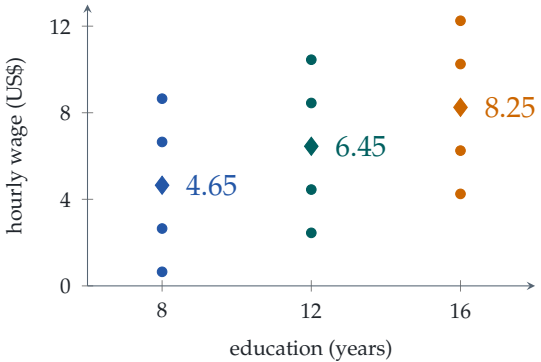


- X: schooling; Y: wage. Five workers at each schooling level.
- $X = 8$: group mean 4.65.
- $X = 12$: group mean 6.45.

Illustrative population; Wooldridge 2.2.1.

Conditional expectation: worked example

Illustrative population — average within each schooling group



- X: schooling; Y: wage. Five workers at each schooling level.
- $X = 8$: group mean 4.65.
- $X = 12$: group mean 6.45.
- $X = 16$: group mean 8.25.

Illustrative population; Wooldridge 2.2.1.

Conditional expectation: worked example

Each group mean is an ordinary average of five equally likely wages

HOURLY WAGES IN DOLLARS; SCHOOLING X IN YEARS

All five workers within each group have equal weight 1/5.

$$\mu(8) = \frac{.65 + 2.65 + 4.65 + 6.65 + 8.65}{5} = 4.65.$$

Conditional expectation: worked example

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$$\mu(12) = \frac{2.45 + 4.45 + 6.45 + 8.45 + 10.45}{5} = 6.45.$$

Conditional expectation: worked example

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All five workers within each group have equal weight 1/5.

$$\mu(8) = \frac{.65 + 2.65 + 4.65 + 6.65 + 8.65}{5} = 4.65.$$

$$\mu(12) = \frac{2.45 + 4.45 + 6.45 + 8.45 + 10.45}{5} = 6.45.$$

$$\mu(16) = \frac{4.25 + 6.25 + 8.25 + 10.25 + 12.25}{5} = 8.25.$$

We have assembled the conditional-mean function, one group at a time.

Conditional expectation: three readings

The same example — a number, a function, and a random variable

Schooling x	8	12	16
Mean hourly wage $\mu(x)$	4.65	6.45	8.25

- **A number**: $E(Y \mid X = 12) = 6.45$.

Conditional expectation: three readings

The same example — a number, a function, and a random variable

Schooling x	8	12	16
Mean hourly wage $\mu(x)$	4.65	6.45	8.25

- **A number**: $E(Y \mid X = 12) = 6.45$.
- **A function**: μ maps each schooling value to its group mean.

Conditional expectation: three readings

The same example — a number, a function, and a random variable

Schooling x	8	12	16
Mean hourly wage $\mu(x)$	4.65	6.45	8.25

- **A number**: $E(Y \mid X = 12) = 6.45$.
- **A function**: μ maps each schooling value to its group mean.
- **A random variable**: select a worker at random. Their X determines which value $\mu(X) = E(Y \mid X)$ takes.

These means describe differences across groups; causality needs more assumptions.

Conditional expectation: the weights

Wooldridge 2.2.1 — summation or integration within the group

For a discrete outcome,

$$E(Y \mid X = x) = \sum_{y} y \, P(Y = y \mid X = x).$$

Conditional expectation: the weights

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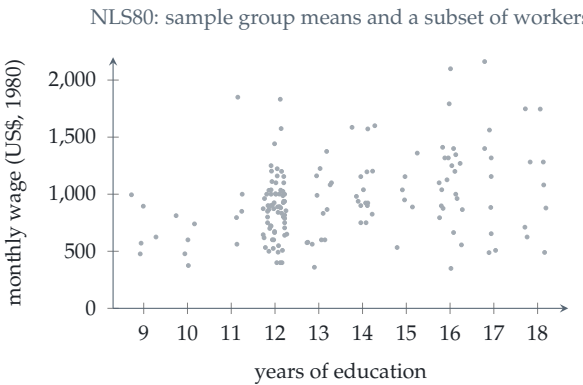
For a continuous outcome,

$$E(Y \mid X = x) = \int y f_{Y|X}(y \mid x) \, dy.$$

Average the possible outcomes using their probabilities within the group.

Conditional expectation: real data

NLS80, 935 men — a model summarises the sample group means

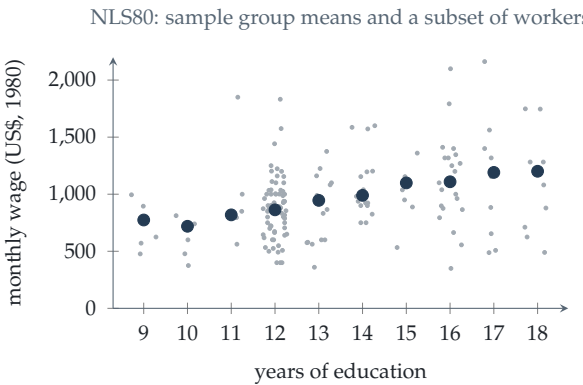


- Grey dots: a subset of individual workers, grouped by schooling.

Wooldridge (2010), Example 4.3 data; course file: `nls80.dta`. All 935 men determine means and fit.

Conditional expectation: real data

NLS80, 935 men — a model summarises the sample group means

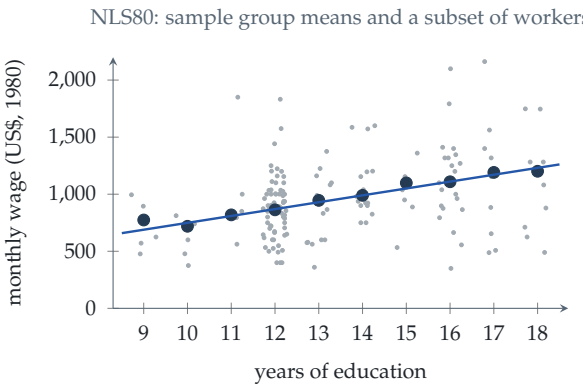


- Grey dots: a subset of individual workers, grouped by schooling.
- Dark dots: sample mean wages at each schooling level.

Wooldridge (2010), Example 4.3 data; course file: `nls80.dta`. All 935 men determine means and fit.

Conditional expectation: real data

NLS80, 935 men — a model summarises the sample group means



- Grey dots: a subset of individual workers, grouped by schooling.
- Dark dots: sample mean wages at each schooling level.
- The fitted line is a simple parametric summary. It need not pass through every mean.

Wooldridge (2010), Example 4.3 data; course file: `nls80.dta`. All 935 men determine means and fit.

Parametric models

Wooldridge 2.2.1, Example 2.1 — models A and B restrict the shape of the CE

A parametric model describes $\mu(x_1, x_2)$ using finitely many coefficients.

MODEL A: LINEAR

$$\mu_A = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Linear in the regressors and in the coefficients.

Parametric models

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A parametric model describes $\mu(x_1, x_2)$ using finitely many coefficients.

MODEL A: LINEAR

$$\mu_A = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Linear in the regressors and in the coefficients.

MODEL B: QUADRATIC

$$\mu_B = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_2^2$$

Curved in x_2, but still linear in the coefficients.

Parametric models

Wooldridge 2.2.1, Example 2.1 — model C is linear in coefficients; model D is nonlinear

MODEL C: INTERACTION

$$\mu_C = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

Linear in each regressor holding the other fixed; jointly nonlinear in (x_1, x_2). Linear in coefficients.

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MODEL D: EXPONENTIAL MEAN

$$\mu_D = \exp[\beta_0 + \beta_1 \log x_1 + \beta_2 x_2]$$

Nonlinear in the coefficients. Suitable for a positive mean; $x_1 > 0$.

Each model answers the same question differently: how does the group mean move when one regressor moves? That is the next section.

SECTION 2 · Wooldridge 2.2.2

Partial effects and elasticities

Partial effects

Wooldridge 2.2.2 — from deterministic to stochastic relationships

- In a deterministic model, $y = f(\mathbf{x})$: the inputs determine the outcome.
- In economic data, workers with the same observed characteristics can earn different wages. This is a stochastic relationship.
- We describe how $E(y \mid \mathbf{x})$ changes as one regressor changes, holding the others fixed.

Partial effects

Wooldridge 2.2.2, (2.6) — a local change in the conditional mean

For a continuous regressor x_j,

$$PE_j(\mathbf{x}) = \frac{\partial \mu(\mathbf{x})}{\partial x_j}.$$

Partial effects

Wooldridge 2.2.2, (2.6) — a local change in the conditional mean

For a continuous regressor x_j,

$$PE_j(\mathbf{x}) = \frac{\partial \mu(\mathbf{x})}{\partial x_j}.$$

$$\Delta E(y \mid \mathbf{x}) \;\approx\; PE_j(\mathbf{x})\,\Delta x_j.$$

- Hold every other regressor fixed.
- Units: units of y per unit of x_j.
- A derivative gives a local rate; finite changes can differ in a curved model.

Partial effects of a discrete regressor

Wooldridge 2.2.2 — finite differences compare two group means

Let $d \in \{0, 1\}$ and let $\mathbf{c}$ contain the other regressors.

$$E(y \mid d = 1, \mathbf{c}) - E(y \mid d = 0, \mathbf{c}).$$

Partial effects of a discrete regressor

Wooldridge 2.2.2 — finite differences compare two group means

Let $d \in \{0, 1\}$ and let $\mathbf{c}$ contain the other regressors.

$$E(y \mid d = 1, \mathbf{c}) - E(y \mid d = 0, \mathbf{c}).$$

- Compare two groups at the same values of $\mathbf{c}$.
- Example: a qualification indicator, with schooling and experience held fixed.
- The difference is in units of y. There is no derivative with respect to the binary indicator.

Partial effects in models A–D

Model A — both slopes are constant

MODEL A

$$\mu_A = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

What are the partial effects of x_1 and x_2?

Partial effects in models A–D

Model A — both slopes are constant

MODEL A

$$\mu_A = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

What are the partial effects of x_1 and x_2?

$$PE_1 = \beta_1, \qquad PE_2 = \beta_2$$

Both partial effects are constant.

Partial effects in models A–D

Model B — the x_2 slope changes with x_2

MODEL B

$$\mu_B = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_2^2$$

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Partial effects in models A–D

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$$\mu_B = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_2^2$$

What are the partial effects of x_1 and x_2?

$$PE_1 = \beta_1, \qquad PE_2 = \beta_2 + 2\beta_3 x_2$$

The x_2 effect depends on the value of x_2.

Partial effects in models A–D

Model C — one regressor changes the other's slope

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What are the partial effects of x_1 and x_2?

$$PE_1 = \beta_1 + \beta_3 x_2, \qquad PE_2 = \beta_2 + \beta_3 x_1$$

Hold the other regressor fixed when taking each derivative.

Partial effects in models A–D

Model D — slopes depend on the conditional mean

MODEL D

$$\mu_D = \exp[\beta_0 + \beta_1 \log x_1 + \beta_2 x_2]$$

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Partial effects in models A–D

Model D — slopes depend on the conditional mean

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$$\mu_D = \exp[\beta_0 + \beta_1 \log x_1 + \beta_2 x_2]$$

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$$PE_1 = \mu_D \frac{\beta_1}{x_1}, \qquad PE_2 = \mu_D \beta_2$$

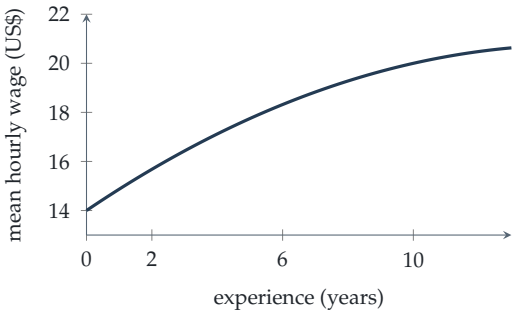
Here μ_D means the entire exponential expression above.

Partial effects: worked example

Hypothetical wage model — the experience slope falls with experience

CONDITIONAL MEAN

$$\mu(x) = 14 + 0.9x - 0.03x^2, \qquad x = \text{years of experience.}$$

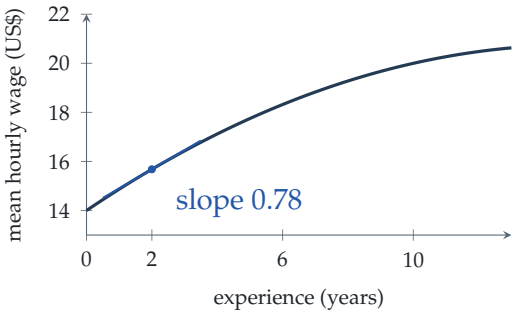


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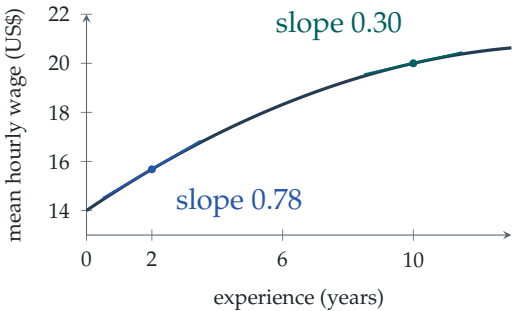
- $PE(x) = 0.9 - 0.06x$.
- At two years: 0.78 dollars per hour, per year of experience.

Partial effects: worked example

Hypothetical wage model — the experience slope falls with experience

CONDITIONAL MEAN

$$\mu(x) = 14 + 0.9x - 0.03x^2, \qquad x = \text{years of experience.}$$



- $PE(x) = 0.9 - 0.06x$.
- At two years: 0.78 dollars per hour, per year of experience.
- At ten years: 0.30 in the same units.

Why elasticities?

Changing units changes the number, not the relationship

Suppose one more year of schooling is associated with a \$60 increase in mean monthly earnings.

Wage units	Partial effect	Economic comparison
Dollars	60	One extra schooling year

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Suppose one more year of schooling is associated with a \$60 increase in mean monthly earnings.

Wage units	Partial effect	Economic comparison
Dollars	60	One extra schooling year
Cents	6,000	The same extra year

Percentage changes remove the arbitrary scale of measurement.

Elasticity uses a percentage change on both sides of the comparison.

Elasticities

Wooldridge 2.2.2, (2.9)–(2.10) — a proportional change in the conditional mean

For a positive conditional mean $\mu(\mathbf{x})$ and $x_j > 0$,

$$\eta_j(\mathbf{x}) = \frac{\partial \mu(\mathbf{x})}{\partial x_j} \frac{x_j}{\mu(\mathbf{x})}.$$

Elasticities

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For a positive conditional mean $\mu(\mathbf{x})$ and $x_j > 0$,

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A 1% increase in x_j is associated with approximately an η_j% change in the conditional mean, holding the other regressors fixed.

Elasticities

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For a positive conditional mean $\mu(\mathbf{x})$ and $x_j > 0$,

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A 1% increase in x_j is associated with approximately an η_j% change in the conditional mean, holding the other regressors fixed.

$$\eta_j(\mathbf{x}) = \frac{\partial \log \mu(\mathbf{x})}{\partial \log x_j}.$$

These are two equivalent expressions for the same elasticity.

Elasticities in models A–D

Model A — $\eta_j = PE_j\, x_j/\mu$; positive mean and regressors

MODEL A

$$\mu_A = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

What are the elasticities with respect to x_1 and x_2?

Elasticities in models A–D

Model A — $\eta_j = PE_j\, x_j/\mu$; positive mean and regressors

MODEL A

$$\mu_A = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

What are the elasticities with respect to x_1 and x_2?

$$\eta_1 = \frac{\beta_1 x_1}{\mu_A}, \qquad \eta_2 = \frac{\beta_2 x_2}{\mu_A}$$

A coefficient can be constant even though an elasticity varies with the covariates.

Elasticities in models A–D

Model B — $\eta_j = PE_j \, x_j/\mu$; positive mean and regressors

MODEL B

$$\mu_B = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_2^2$$

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Elasticities in models A–D

Model B — $\eta_j = PE_j\, x_j/\mu$; positive mean and regressors

MODEL B

$$\mu_B = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_2^2$$

What are the elasticities with respect to x_1 and x_2?

$$\eta_1 = \frac{\beta_1 x_1}{\mu_B}, \qquad \eta_2 = \frac{(\beta_2 + 2\beta_3 x_2) x_2}{\mu_B}$$

A coefficient can be constant even though an elasticity varies with the covariates.

Elasticities in models A–D

Model C — $\eta_j = PE_j\, x_j/\mu$; positive mean and regressors

MODEL C

$$\mu_C = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

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Elasticities in models A–D

Model C — $\eta_j = PE_j\, x_j/\mu$; positive mean and regressors

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$$\mu_C = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

What are the elasticities with respect to x_1 and x_2?

$$\eta_1 = \frac{(\beta_1 + \beta_3 x_2)x_1}{\mu_C}, \qquad \eta_2 = \frac{(\beta_2 + \beta_3 x_1)x_2}{\mu_C}$$

A coefficient can be constant even though an elasticity varies with the covariates.

Elasticities in models A–D

Model D — $\eta_j = PE_j\, x_j/\mu$; positive mean and regressors

MODEL D

$$\mu_D = \exp[\beta_0 + \beta_1 \log x_1 + \beta_2 x_2]$$

What are the elasticities with respect to x_1 and x_2?

Elasticities in models A–D

Model D — $\eta_j = PE_j\, x_j/\mu$; positive mean and regressors

MODEL D

$$\mu_D = \exp[\beta_0 + \beta_1 \log x_1 + \beta_2 x_2]$$

What are the elasticities with respect to x_1 and x_2?

$$\eta_1 = \beta_1, \qquad \eta_2 = \beta_2 x_2$$

The x_1 elasticity is constant; the x_2 elasticity generally is not.

Elasticities: real data

NLS80 — a log–log slope is a percentage comparison

ILLUSTRATIVE FITTED REGRESSION; 935 MEN

$$\widehat{\log(wage)} = \widehat{a} + 0.83 \log(educ).$$

- Take logs of both variables, then fit the regression.

Elasticities: real data

NLS80 — a log–log slope is a percentage comparison

ILLUSTRATIVE FITTED REGRESSION; 935 MEN

$$\widehat{\log(wage)} = \hat{a} + 0.83 \log(educ).$$

- Take logs of both variables, then fit the regression.
- A 1% increase in schooling is associated with about a 0.83% increase in the fitted geometric mean wage.

Wooldridge Example 4.3 data; `nls80.dta`. Bivariate illustration, not a causal estimate.

Semielasticities

Wooldridge 2.2.2, (2.13)–(2.14) — percent change in the mean per unit of x_j

For $\mu(\mathbf{x}) = E(Y \mid \mathbf{x}) > 0$, the arithmetic-mean semielasticity is:

$$s_j(\mathbf{x}) = \frac{\partial \log \mu(\mathbf{x})}{\partial x_j}.$$

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$$s_j(\mathbf{x}) = \frac{\partial \log \mu(\mathbf{x})}{\partial x_j}.$$

- A one-unit increase in x_j changes the arithmetic conditional mean by approximately $100 s_j$ percent.
- For a log-wage fit, $\log(wage) = \beta_0 + \beta_1\, educ + \beta_2\, exper + u$. Here $100\beta_1$ is the approximate percent change in fitted geometric mean wage per extra schooling year.

Mean-log and log-mean slopes can differ. Their relation is in Appendix A. In model D, $s_2 = \beta_2$.

▸ Appendix: two elasticity definitions

Three slopes, three interpretations

NLS80 — level–level, log–level and log–log, all on schooling

Let $\mu(X) = E(Y \mid X)$ and $\ell(X) = E(\log Y \mid X)$.

Model	Slope definition	Fitted wage interpretation
Level–level $\mu = a + bX$	$b = \partial\mu/\partial X$	Dollars/month per year: $b = 60.2$: +1 year $\Rightarrow$ about +\$60/month

NLS80 bivariate fits. Log rows describe fitted geometric mean wage. X: schooling (years); Y: monthly earnings (dollars).

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Log–level $\ell = a + bX$	$b = \partial\ell/\partial X$	Percent per year: $b = .060$: +1 year ⇒ about +6%

NLS80 bivariate fits. Log rows describe fitted geometric mean wage. X: schooling (years); Y: monthly earnings (dollars).

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Log–level $\ell = a + bX$	$b = \partial\ell/\partial X$	Percent per year: $b = .060$: +1 year ⇒ about +6%
Log–log $\ell = a + b \log X$	$b = \partial\ell/\partial \log X$	Percent per 1%: $b = .83$: +1% schooling ⇒ about +.83%

NLS80 bivariate fits. Log rows describe fitted geometric mean wage. X: schooling (years); Y: monthly earnings (dollars).

SECTION 3 · Wooldridge 2.2.3

The error form

The conditional-expectation error

Wooldridge 2.2.3, (2.15)–(2.16) — a decomposition that always holds

We have described group averages. Individual outcomes still vary.

$$Y = \underbrace{E(Y \mid \mathbf{X})}_{\text{group mean}} + \underbrace{u}_{\text{individual deviation}} .$$

The conditional-expectation error

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$$Y = \underbrace{E(Y \mid \mathbf{X})}_{\text{group mean}} + \underbrace{u}_{\text{individual deviation}} .$$

$$u \equiv Y - E(Y \mid \mathbf{X}), \qquad E(u \mid \mathbf{X}) = 0.$$

Subtract each group's true mean: its deviations average to zero.

The conditional-expectation error

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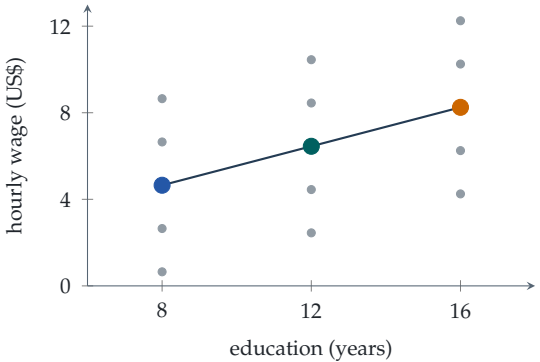
Subtract each group's true mean: its deviations average to zero.

The CE error has zero conditional mean by definition.

▸ Why $E(u \mid \mathbf{X}) = 0$ follows from the definition

The CE error: worked example

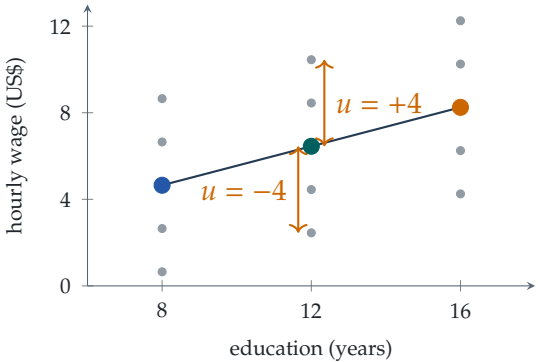
Illustrative population — errors average to zero within every group



- Schooling groups: 8, 12, 16 years. Means: 4.65, 6.45, 8.25.

The CE error: worked example

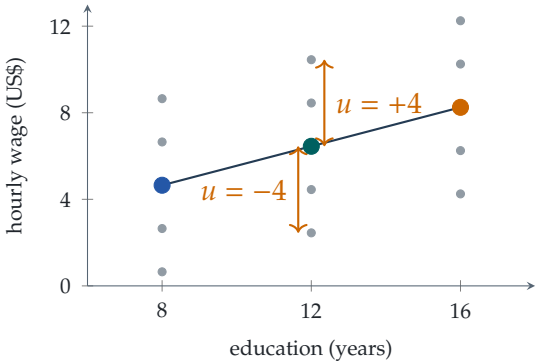
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- $u = Y - E(Y \mid X)$ is a worker's vertical distance from the group mean.

The CE error: worked example

Illustrative population — errors average to zero within every group



- Schooling groups: 8, 12, 16 years. Means: 4.65, 6.45, 8.25.
- $u = Y - E(Y \mid X)$ is a worker's vertical distance from the group mean.
- Every group's errors are $-4, -2, 0, 2, 4$.

$$E(u \mid X) = 0.$$

The error assumption: house prices

Wooldridge Example 2.2 — what does the distance coefficient measure?

$$hprice = \beta_0 + \beta_1\, sqrft + \beta_2\, distance + u.$$

- *hprice*: price in thousands of dollars; *sqrft*: floor area in square feet.
- *distance*: kilometres from a city incinerator. These units are chosen for this illustration.

If the linear part is the CE, what does β_2 measure: a partial effect, an elasticity, or a semielasticity?

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If the linear part is the CE, what does β_2 measure: a partial effect, an elasticity, or a semielasticity?

A partial effect: β_2 thousand dollars per additional kilometre, at fixed floor area.

The error assumption: house prices

The error mean can change the distance slope

PRICE: THOUSAND DOLLARS; DISTANCE: KM; FLOOR AREA: SQUARE FEET

$$hprice = \beta_0 + \beta_1 \, sqrft + \beta_2 \, distance + u.$$

The error assumption: house prices

The error mean can change the distance slope

PRICE: THOUSAND DOLLARS; DISTANCE: KM; FLOOR AREA: SQUARE FEET

$$hprice = \beta_0 + \beta_1\, sqrft + \beta_2\, distance + u.$$

$$E(hprice \mid sqrft, distance) = \beta_0 + \beta_1\, sqrft + \beta_2\, distance + E(u \mid sqrft, distance).$$

The error assumption: house prices

The error mean can change the distance slope

PRICE: THOUSAND DOLLARS; DISTANCE: KM; FLOOR AREA: SQUARE FEET

$$hprice = \beta_0 + \beta_1\, sqrft + \beta_2\, distance + u.$$

$$E(hprice \mid sqrft, distance) = \beta_0 + \beta_1\, sqrft + \beta_2\, distance + E(u \mid sqrft, distance).$$

$$\frac{\partial E(hprice \mid sqrft, distance)}{\partial distance} = \beta_2 + \frac{\partial E(u \mid sqrft, distance)}{\partial distance}.$$

The observed mean-price slope can include a systematic change in omitted factors.

The error assumption: house prices

What must $E(u \mid sqrft, distance) = 0$ rule out?

THE PROPOSED MODEL

$$hprice = \beta_0 + \beta_1 sqrft + \beta_2 distance + u.$$

What could be in u, and could its average change with distance even among houses of the same size?

The error assumption: house prices

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- Neighbourhood quality or housing age could change systematically with distance.
- A sufficient model condition is $E(u \mid sqrft, distance) = 0$.

The error assumption: house prices

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What could be in u, and could its average change with distance even among houses of the same size?

- Neighbourhood quality or housing age could change systematically with distance.
- A sufficient model condition is $E(u \mid sqrft, distance) = 0$.
- Causality also needs a credible economic justification for the restriction and controls.

SECTION 4 · Wooldridge 2.2.4

Properties of conditional expectations and LIE

Properties of conditional expectations

Wooldridge 2.2.4, App. 2A — known functions pass through; the operator is linear

A known function

$$E[g(\mathbf{x}) \mid \mathbf{x}] = g(\mathbf{x}).$$

Properties of conditional expectations

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Once schooling is known, its square is known too: nothing is left to average.

Properties of conditional expectations

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Linearity

$$E(aY + bZ \mid \mathbf{x}) = aE(Y \mid \mathbf{x}) + bE(Z \mid \mathbf{x}).$$

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For constants a, b, averaging a sum gives the sum of the averages.

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Linearity

$$E(aY + bZ \mid \mathbf{x}) = aE(Y \mid \mathbf{x}) + bE(Z \mid \mathbf{x}).$$

For constants a, b, averaging a sum gives the sum of the averages.

Next, the rule that matters most: averaging over information we lack.

▸ More: taking out what is known ▸ More: transformed regressors

Law of iterated expectations (LIE)

Wooldridge 2.2.4, (2.19); App. 2A, CE.2 — unconditional form: average within groups, then across groups

Y: outcome; X: grouping variable.

1. **Inner average**: find each group's mean, $E(Y \mid X)$.

Law of iterated expectations; Stat 110 calls this form Adam's law. $E(|Y|) < \infty$; use all groups and their correct probabilities. Independence is not required.

Law of iterated expectations (LIE)

Wooldridge 2.2.4, (2.19); App. 2A, CE.2 — unconditional form: average within groups, then across groups

Y: outcome; X: grouping variable.

1. **Inner average**: find each group's mean, $E(Y \mid X)$.
2. **Outer average**: combine those means using the group probabilities.

$$\underbrace{E[\,E(Y \mid X)\,]}_{\text{average of group means}} = \underbrace{E(Y)}_{\text{overall mean}}$$

Law of iterated expectations; Stat 110 calls this form Adam's law. $E(|Y|) < \infty$; use all groups and their correct probabilities. Independence is not required.

Law of iterated expectations: the bottle story

HarvardX, Stat 110 — what can a bottle label tell us?



Every label says: *Contains 200 mg of Bayesium.*
Sources: Discretown, Continuopolis and/or Bayesville.

Fictional example adapted from HarvardX (Blitzstein), Understanding Conditional Expectation. Y: bottle content (mg); X: source city.

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Discretown	**Continuopolis**	**Bayesville**
mean **160 mg**	mean **190 mg**	mean **250 mg**

Fictional example adapted from HarvardX (Blitzstein), Understanding Conditional Expectation. Y: bottle content (mg); X: source city.

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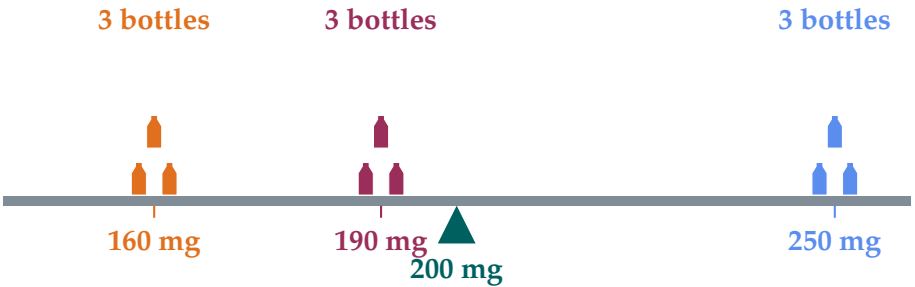
Discretown	**Continuopolis**	**Bayesville**
mean **160 mg**	mean **190 mg**	mean **250 mg**

For a random bottle, what content do you expect?
A) 200 mg B) more C) less D) it depends

Fictional example adapted from HarvardX (Blitzstein), Understanding Conditional Expectation. Y: bottle content (mg); X: source city.

Law of iterated expectations: the bottle story

From equal counts to the actual bottle counts

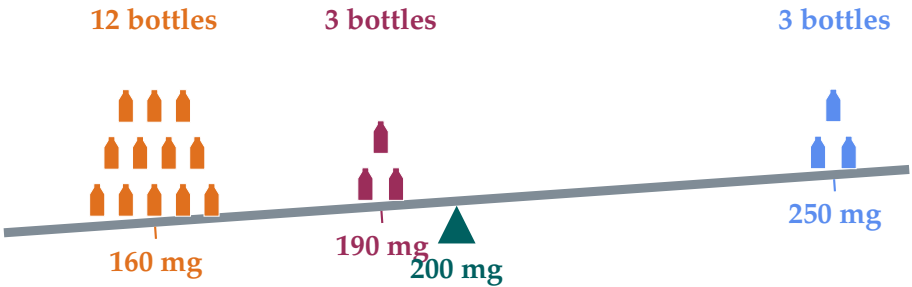


$$\frac{160 + 190 + 250}{3} = 200 \text{ mg}.$$

Equal counts give equal probability weights.

Law of iterated expectations: the bottle story

From equal counts to the actual bottle counts

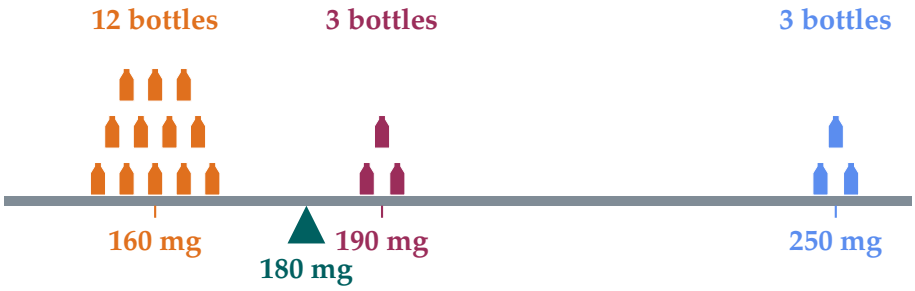


Actual counts: 12, 3 and 3 bottles. Which way must the balance point move?

Each bottle is equally likely; the heavier left stack moves down.

Law of iterated expectations: the bottle story

From equal counts to the actual bottle counts



$$\frac{12}{18} \times 160 + \frac{3}{18} \times 190 + \frac{3}{18} \times 250 = 180 \text{ mg}.$$

Bottle counts determine the city probability weights.

Law of iterated expectations: the bottle story

The balance calculation becomes the law of iterated expectations

X is the city; Y is the content of a randomly selected bottle.

	Discretown	Continuopolis	Bayesville
City mean $E(Y \mid X = x)$	160	190	250

Wooldridge App. 2A, CE.2; Stat 110 calls this Adam's law.

Law of iterated expectations: the bottle story

The balance calculation becomes the law of iterated expectations

X is the city; Y is the content of a randomly selected bottle.

	Discretown	Continuopolis	Bayesville
City mean $E(Y \mid X = x)$	160	190	250
Weight $P(X = x)$	2/3	1/6	1/6

$$E(Y) = \frac{2}{3} \times 160 + \frac{1}{6} \times 190 + \frac{1}{6} \times 250 = 180.$$

Wooldridge App. 2A, CE.2; Stat 110 calls this Adam's law.

Law of iterated expectations: the bottle story

The balance calculation becomes the law of iterated expectations

X is the city; Y is the content of a randomly selected bottle.

	Discretown	Continuopolis	Bayesville
City mean $E(Y \mid X = x)$	160	190	250
Weight $P(X = x)$	2/3	1/6	1/6

$$E(Y) = \sum_x P(X = x)E(Y \mid X = x) = E[\, E(Y \mid X)\,].$$

▸ Appendix D: continue with variance ▸ Appendix C: regrouping people

Wooldridge App. 2A, CE.2; Stat 110 calls this Adam's law.

Law of iterated expectations (LIE)

Wooldridge (2.20)–(2.21), CE.3 — conditional LIE inside a fixed group

Adding Z makes (X, Z) finer information than X alone:

$$E[\,E(Y \mid X, Z) \mid X\,] = E(Y \mid X).$$

Law of iterated expectations (LIE)

Wooldridge (2.20)–(2.21), CE.3 — conditional LIE inside a fixed group

Adding Z makes (X, Z) finer information than X alone:

$$E[\,E(Y \mid X, Z) \mid X\,] = E(Y \mid X).$$

- **Inner**: means within (X, Z) groups. **Outer**: average over Z *within* each X group: the weights are conditional on X.
- The nested, less informative set X remains; Z is averaged out with its distribution *given* X.

The next example starts with all eight people, then applies the same rule within the six women.

▸ Rule: which information set survives?

LIE: worked example

Unconditional mean — start with the full men-and-women population

Draw one of these eight people with equal probability. Y: IQ; X: sex (M or F).

Men (M): 2

A	B
80	90

Women (F): 6

C	D	E	F	G	H
90	100	100	110	110	120

Find the mean for men, the mean for women, and the overall mean.

Invented IQ scores and memberships. $E(Y \mid M)$ and $E(Y \mid F)$ abbreviate conditioning on sex.

LIE: worked example

Unconditional mean — start with the full men-and-women population

Draw one of these eight people with equal probability. Y: IQ; X: sex (M or F).

Men (M): 2

A	B
80	90

Mean IQ: 85

Women (F): 6

C	D	E	F	G	H
90	100	100	110	110	120

Mean IQ: 105

$$E(Y) = E[\,E(Y \mid X)\,] = \tfrac{2}{8}E(Y \mid M) + \tfrac{6}{8}E(Y \mid F)$$

Invented IQ scores and memberships. $E(Y \mid M)$ and $E(Y \mid F)$ abbreviate conditioning on sex.

LIE: worked example

Unconditional mean — start with the full men-and-women population

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Men (M): 2

A	B
80	90

Mean IQ: 85

Women (F): 6

C	D	E	F	G	H
90	100	100	110	110	120

Mean IQ: 105

$$E(Y) = E[\,E(Y \mid X)\,] = \tfrac{2}{8}E(Y \mid M) + \tfrac{6}{8}E(Y \mid F)$$
$$= \tfrac{2}{8} \times 85 + \tfrac{6}{8} \times 105 = 100.$$

Now keep only the six women: the target becomes $E(Y \mid F) = 105$.

Invented IQ scores and memberships. $E(Y \mid M)$ and $E(Y \mid F)$ abbreviate conditioning on sex.

Conditional (nested) LIE: worked example

Within women — split by smoking status, then average the subgroup means

The same six women: $E(Y \mid F) = 105$

$Z = S$: **2 smokers**		$Z = NS$: **4 nonsmokers**			
C	D	E	F	G	H
90	100	100	110	110	120

Find the smoker and nonsmoker mean IQs among these women.

Invented data; adapted from Ben Lambert, Nested LIE. Blue: group kept fixed; orange: subgroup averaged over.

Conditional (nested) LIE: worked example

Within women — split by smoking status, then average the subgroup means

The same six women: $E(Y \mid F) = 105$

$Z = S$: **2 smokers**		$Z = NS$: **4 nonsmokers**			
C	D	E	F	G	H
90	100	100	110	110	120
Mean IQ: 95		Mean IQ: 110			

How can those two subgroup means recover 105? Which population supplies the weights?

Invented data; adapted from Ben Lambert, Nested LIE. Blue: group kept fixed; orange: subgroup averaged over.

Conditional (nested) LIE: worked example

Within women — split by smoking status, then average the subgroup means

The same six women: $E(Y \mid F) = 105$

$Z = S$: **2 smokers**		$Z = NS$: **4 nonsmokers**			
C	D	E	F	G	H
90	100	100	110	110	120
Mean IQ: 95		Mean IQ: 110			

$$E(Y \mid F) = E[\, E(Y \mid F, Z) \mid F \,]$$
$$= P(Z = S \mid F)\, E(Y \mid F, Z = S) + P(Z = NS \mid F)\, E(Y \mid F, Z = NS)$$

Invented data; adapted from Ben Lambert, Nested LIE. Blue: group kept fixed; orange: subgroup averaged over.

Conditional (nested) LIE: worked example

Within women — split by smoking status, then average the subgroup means

The same six women: $E(Y \mid F) = 105$

$Z = S$: **2 smokers**		$Z = NS$: **4 nonsmokers**			
C	D	E	F	G	H
90	100	100	110	110	120
Mean IQ: 95		Mean IQ: 110			

$$
\begin{aligned}
E(Y \mid F) &= E[\, E(Y \mid F, Z) \mid F \,] \\
&= P(Z = S \mid F)\, E(Y \mid F, Z = S) + P(Z = NS \mid F)\, E(Y \mid F, Z = NS) \\
&= \tfrac{2}{6} \times 95 + \tfrac{4}{6} \times 110 = 105.
\end{aligned}
$$

Average over smoking status; keep "female" in every mean and weight.

Invented data; adapted from Ben Lambert, Nested LIE. Blue: group kept fixed; orange: subgroup averaged over.

SECTION 5 · Wooldridge 2.2.4–2.2.5

Average partial effects, unobserved regressors and proxies

Average partial effects (APE)

Wooldridge 2.2.5 — average the effect over unobserved heterogeneity q

- Return to wages: $\mathbf{x}$ is schooling and experience; q is ability, an example of unobserved heterogeneity. Consider $E(y \mid \mathbf{x}, q)$.

Average partial effects (APE)

Wooldridge 2.2.5 — average the effect over unobserved heterogeneity q

- Return to wages: $\mathbf{x}$ is schooling and experience; q is ability, an example of unobserved heterogeneity. Consider $E(y \mid \mathbf{x}, q)$.
- The partial effect $\partial E(y \mid \mathbf{x}, q)/\partial x_j$ generally depends on q.

Average partial effects (APE)

Wooldridge 2.2.5 — average the effect over unobserved heterogeneity q

- Return to wages: $\mathbf{x}$ is schooling and experience; q is ability, an example of unobserved heterogeneity. Consider $E(y \mid \mathbf{x}, q)$.
- The partial effect $\partial E(y \mid \mathbf{x}, q)/\partial x_j$ generally depends on q.
- Usually of more interest: the effect averaged across the population distribution of q, at fixed $\mathbf{x}$, the average partial effect (APE):

$$APE_j(\mathbf{x}) = E_q\left[\frac{\partial E(y \mid \mathbf{x}, q)}{\partial x_j}\right].$$

Average partial effects (APE)

Wooldridge 2.2.5 — average the effect over unobserved heterogeneity q

- Return to wages: $\mathbf{x}$ is schooling and experience; q is ability, an example of unobserved heterogeneity. Consider $E(y \mid \mathbf{x}, q)$.
- The partial effect $\partial E(y \mid \mathbf{x}, q)/\partial x_j$ generally depends on q.
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$$APE_j(\mathbf{x}) = E_q\left[\frac{\partial E(y \mid \mathbf{x}, q)}{\partial x_j}\right].$$

- Can we learn the APE from CEs that involve only observed variables?

Wages: $\mathbf{x}$ is schooling and experience, q ability. q is averaged with its population weights; ability and the covariates may be related.

Unobserved regressor

Wooldridge 2.2.4 — the target CE and the CE we can actually study

TARGET (TRUE MODEL): s SCHOOLING, e EXPERIENCE, q UNOBSERVED ABILITY

$$E(y \mid s, e, q) = \beta_0 + \beta_s s + \beta_e e + \gamma q.$$

- We observe y, s, e but not q. The data can reveal the CE on the *observed* variables. For this illustration, assume it is linear: $E(y \mid s, e) = \alpha_0 + \alpha_s s + \alpha_e e$.

Unobserved regressor

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TARGET (TRUE MODEL): s SCHOOLING, e EXPERIENCE, q UNOBSERVED ABILITY

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- We observe y, s, e but not q. The data can reveal the CE on the *observed* variables. For this illustration, assume it is linear: $E(y \mid s, e) = \alpha_0 + \alpha_s s + \alpha_e e$.
- Three objects: target β; observed α; and, next, the auxiliary link between ability and the regressors.

What do the α's measure? Must α_s equal β_s?

Unobserved regressor: derivation

Wooldridge (2.24) — LIE replaces missing ability by its group average

STRUCTURAL CE; ABILITY q IS UNOBSERVED. TARGET SLOPES IN BLUE, THE AVERAGED ABILITY TERM IN ORANGE

$$E(y \mid s, e, q) = \beta_0 + \beta_s s + \beta_e e + \gamma q.$$

Wooldridge (2.24): "by the LIE, and the linearity of the CE operator". Properties used: App. 2A, CE.3, then CE.1.

▸ Why LIE is needed: the error route

Unobserved regressor: derivation

Wooldridge (2.24) — LIE replaces missing ability by its group average

STRUCTURAL CE; ABILITY q IS UNOBSERVED. TARGET SLOPES IN BLUE, THE AVERAGED ABILITY TERM IN ORANGE

$$E(y \mid s, e, q) = \beta_0 + \beta_s s + \beta_e e + \gamma q.$$

$$E(y \mid s, e) = E[\, E(y \mid s, e, q) \mid s, e \,] \qquad \text{conditional LIE}$$

Wooldridge (2.24): "by the LIE, and the linearity of the CE operator". Properties used: App. 2A, CE.3, then CE.1.

▸ Why LIE is needed: the error route

Unobserved regressor: derivation

Wooldridge (2.24) — LIE replaces missing ability by its group average

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$$\begin{aligned} E(y \mid s, e) &= E[\, E(y \mid s, e, q) \mid s, e \,] && \text{conditional LIE} \\ &= E[\, \beta_0 + \beta_s s + \beta_e e + \gamma q \mid s, e \,] && \text{substitute the structural CE} \end{aligned}$$

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$$\begin{aligned} E(y \mid s, e) &= E[\, E(y \mid s, e, q) \mid s, e \,] && \text{conditional LIE} \\ &= E[\, \beta_0 + \beta_s s + \beta_e e + \gamma q \mid s, e \,] && \text{substitute the structural CE} \\ &= \beta_0 + \beta_s s + \beta_e e + \gamma\, E(q \mid s, e). && \text{linearity; } s, e \text{ are known} \end{aligned}$$

Among people with a particular schooling and experience level, replace ability by that group's average ability.

Wooldridge (2.24): "by the LIE, and the linearity of the CE operator". Properties used: App. 2A, CE.3, then CE.1.

▸ Why LIE is needed: the error route

Unobserved regressor: derivation

Substitute the average ability in each schooling–experience group

FROM LIE

$$E(y \mid s, e) = \beta_0 + \beta_s s + \beta_e e + \gamma E(q \mid s, e).$$

Suppose the auxiliary mean is

$$E(q \mid s, e) = \delta_0 + \delta_s s + \delta_e e.$$

Unobserved regressor: derivation

Substitute the average ability in each schooling–experience group

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$$E(y \mid s, e) = \beta_0 + \beta_s s + \beta_e e + \gamma E(q \mid s, e).$$

Suppose the auxiliary mean is

$$E(q \mid s, e) = \delta_0 + \delta_s s + \delta_e e.$$

$$E(y \mid s, e) = \beta_0 + \beta_s s + \beta_e e + \gamma(\delta_0 + \delta_s s + \delta_e e).$$

The last term can change with schooling and experience. Now collect like terms.

Unobserved regressor: composite slopes

Observed slopes combine the target and the auxiliary relationship

AFTER SUBSTITUTION: TARGET SLOPES IN BLUE, AUXILIARY PARAMETERS IN ORANGE, COMPOSITE (OBSERVED) COEFFICIENTS IN RED

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$$\alpha_0 = \beta_0 + \gamma\delta_0, \quad \alpha_s = \beta_s + \gamma\delta_s, \quad \alpha_e = \beta_e + \gamma\delta_e.$$

Unobserved regressor: composite slopes

Observed slopes combine the target and the auxiliary relationship

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$$\alpha_0 = \beta_0 + \gamma\delta_0, \quad \alpha_s = \beta_s + \gamma\delta_s, \quad \alpha_e = \beta_e + \gamma\delta_e.$$

These composite coefficients describe the observed CE. The schooling slope equals its target when $\gamma\delta_s = 0$.

▸ Real data: adding IQ ▸ Omitted interactions

Wrap-up

ECO6102 / Lecture 2

Key takeaways

What we defined, assumed and learned

- **Define**: the conditional mean and its error; the best linear predictor waits in the appendix.
- **Interpret**: partial effects, elasticities, semielasticities and explicitly defined averages of effects.
- **Relate**: LIE connects finer and coarser information sets.
- **Assume**: correct specification and credible identifying restrictions are needed for the intended economic interpretation.
- **Recover**: suitable proxies can recover target slopes; an arbitrary proxy can leave bias.

Next lecture

Lecture 3 builds on today's definitions

- Today: Conditional Expectations — Wooldridge (2010), Chapter 2.
- Next: Linear Regression, Wooldridge (2010), Chapter 4 (Weeks 3–4).
- OLS estimates a linear projection. We will study when its coefficients have the economic interpretation we want. ▸ Appendix: linear projections
- No lecture on Monday 5 October 2026; Instrumental Variables begins the following Monday.

Adapted from slides by Shuyang Sheng and Qihui Chen

Appendix

Appendix contents

Reached from the buttons in the main lecture; every frame has a return button

A	Elasticities: two definitions	App. 1
B	The error form	App. 4
C	More on conditional expectations and LIE	App. 6
D	Variance decomposition: the bottle story continued	App. 15
E	Omitted variables and proxies	App. 18
F	Linear projections	App. 23

APPENDIX A · Wooldridge 2.2.2

Elasticities: two definitions

Elasticities: two definitions

Wooldridge 2.2.2 — logging before averaging changes the object

Elasticity of the arithmetic conditional mean

$$\eta_j = \frac{\partial \log E(Y \mid \mathbf{x})}{\partial \log x_j}$$

Elasticities: two definitions

Wooldridge 2.2.2 — logging before averaging changes the object

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Alternative based on the conditional mean of log wages

$$\widetilde{\eta}_j = \frac{\partial E(\log Y \mid \mathbf{x})}{\partial \log x_j}$$

Elasticities: two definitions

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Alternative based on the conditional mean of log wages

$$\widetilde{\eta}_j = \frac{\partial E(\log Y \mid \mathbf{x})}{\partial \log x_j}$$

The second requires $Y > 0$. The first can apply when some outcomes are zero but their conditional mean is positive.

Back to main lecture

Elasticities: two definitions

Two equally likely wages — taking logs and averaging do not commute

Average, then take logs

$$E(Y) = \tfrac{4+16}{2} = 10$$

$$\log E(Y) = \log 10 \approx 2.30$$

Wooldridge 2.2.2, p. 17; Appendix 2A, CE.7.

Back to main lecture

Elasticities: two definitions

Two equally likely wages — taking logs and averaging do not commute

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$$E(Y) = \tfrac{4+16}{2} = 10$$
$$\log E(Y) = \log 10 \approx 2.30$$

Take logs, then average

$$E(\log Y) = \tfrac{\log 4 + \log 16}{2}$$
$$= \log 8 \approx 2.08$$

Wooldridge 2.2.2, p. 17; Appendix 2A, CE.7.

Back to main lecture

Elasticities: two definitions

Two equally likely wages — taking logs and averaging do not commute

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Take logs, then average

$$E(\log Y) = \tfrac{\log 4 + \log 16}{2}$$
$$= \log 8 \approx 2.08$$

Exponentiating the mean log gives the geometric mean ($8), not the arithmetic mean ($10).

Jensen: $E(\log Y) \leq \log E(Y)$, also within each group.

Wooldridge 2.2.2, p. 17; Appendix 2A, CE.7.

Back to main lecture

Elasticities: two definitions

A sufficient condition for the two to agree in a log-outcome model

LOG–LOG IN x_1, LOG–LEVEL IN x_2

$$\log Y = \beta_0 + \beta_1 \log x_1 + \beta_2 x_2 + u.$$

- If $E(u \mid \mathbf{x}) = 0$, the mean-log definition gives $\widetilde{\eta}_1 = \beta_1$.

Elasticities: two definitions

A sufficient condition for the two to agree in a log-outcome model

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- If $E(u \mid \mathbf{x}) = 0$, the mean-log definition gives $\widetilde{\eta}_1 = \beta_1$.
- If u is also independent of $\mathbf{x}$, with $E(e^u) < \infty$, both definitions give $\eta_1 = \widetilde{\eta}_1 = \beta_1$.

Elasticities: two definitions

A sufficient condition for the two to agree in a log-outcome model

LOG–LOG IN x_1, LOG–LEVEL IN x_2

$$\log Y = \beta_0 + \beta_1 \log x_1 + \beta_2 x_2 + u.$$

- If $E(u \mid \mathbf{x}) = 0$, the mean-log definition gives $\widetilde{\eta}_1 = \beta_1$.
- If u is also independent of $\mathbf{x}$, with $E(e^u) < \infty$, both definitions give $\eta_1 = \widetilde{\eta}_1 = \beta_1$.
- Independence makes the adjustment from the geometric mean to the arithmetic mean constant across $\mathbf{x}$.

Back to main lecture

APPENDIX B · Wooldridge 2.2.3

The error form

The conditional-expectation error

Why $E(u \mid \mathbf{X}) = 0$ follows from the definition, not from an assumption

Definition: $u \equiv Y - E(Y \mid \mathbf{X})$. Take the conditional expectation of both sides, given $\mathbf{X}$:

Wooldridge 2.2.3, (2.15)–(2.16); rules CE.1 and known-function, App. 2A; properties in the main lecture.

Back to main lecture

The conditional-expectation error

Why $E(u \mid \mathbf{X}) = 0$ follows from the definition, not from an assumption

Definition: $u \equiv Y - E(Y \mid \mathbf{X})$. Take the conditional expectation of both sides, given $\mathbf{X}$:

$$E(u \mid \mathbf{X}) = E\Big[\, Y - E(Y \mid \mathbf{X}) \,\Big|\, \mathbf{X} \Big]$$

Wooldridge 2.2.3, (2.15)–(2.16); rules CE.1 and known-function, App. 2A; properties in the main lecture.

Back to main lecture

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$$E(u \mid \mathbf{X}) = E\Big[\, Y - E(Y \mid \mathbf{X}) \,\Big|\, \mathbf{X}\Big]$$

$$= E(Y \mid \mathbf{X}) - E\Big[\, E(Y \mid \mathbf{X}) \,\Big|\, \mathbf{X}\Big] \quad \text{linearity of the CE operator}$$

Wooldridge 2.2.3, (2.15)–(2.16); rules CE.1 and known-function, App. 2A; properties in the main lecture.

Back to main lecture

The conditional-expectation error

Why $E(u \mid \mathbf{X}) = 0$ follows from the definition, not from an assumption

Definition: $u \equiv Y - E(Y \mid \mathbf{X})$. Take the conditional expectation of both sides, given $\mathbf{X}$:

$$
\begin{aligned}
E(u \mid \mathbf{X}) &= E\Big[\, Y - E(Y \mid \mathbf{X}) \,\Big|\, \mathbf{X} \Big] \\
&= E(Y \mid \mathbf{X}) - E\Big[\, E(Y \mid \mathbf{X}) \,\Big|\, \mathbf{X} \Big] && \text{linearity of the CE operator} \\
&= E(Y \mid \mathbf{X}) - E(Y \mid \mathbf{X}) = 0. && E(Y \mid \mathbf{X}) \text{ is a function of } \mathbf{X}\text{: already known}
\end{aligned}
$$

By construction; a *chosen* model earns it only if it equals the true CE (next frame).

Wooldridge 2.2.3, (2.15)–(2.16); rules CE.1 and known-function, App. 2A; properties in the main lecture.

Back to main lecture

Misspecification

Wooldridge 2.2.3 — a chosen model can leave systematic errors

Let $\mu(\mathbf{x})$ be the true CE and $m(\mathbf{x})$ a proposed model.

$$v = Y - m(\mathbf{X}) = \underbrace{Y - \mu(\mathbf{X})}_{\text{CE error}} + \underbrace{\mu(\mathbf{X}) - m(\mathbf{X})}_{\text{misspecification}}.$$

Misspecification

Wooldridge 2.2.3 — a chosen model can leave systematic errors

Let $\mu(\mathbf{x})$ be the true CE and $m(\mathbf{x})$ a proposed model.

$$v = Y - m(\mathbf{X}) = \underbrace{Y - \mu(\mathbf{X})}_{\text{CE error}} + \underbrace{\mu(\mathbf{X}) - m(\mathbf{X})}_{\text{misspecification}}.$$

So $E(v \mid \mathbf{X}) = \mu(\mathbf{X}) - m(\mathbf{X})$:

- Zero conditional mean requires the proposed model to equal the true CE on the population's support.
- A linear model can miss curvature or interactions in group averages.

Back to main lecture

APPENDIX C · Wooldridge 2.2.4

More on conditional expectations and LIE

LIE: worked example

Eight equally likely people — one population, one overall mean

Let Y be the IQ score of a randomly selected person.

Person	A	B	C	D	E	F	G	H
IQ score	80	90	90	100	100	110	110	120

What is the expected IQ? Could we get it by averaging within groups first?

Invented population. Partition intuition: Ben Lambert, The Law of Iterated Expectations: an introduction.

Back to main lecture

LIE: worked example

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Person	A	B	C	D	E	F	G	H
IQ score	80	90	90	100	100	110	110	120

What is the expected IQ? Could we get it by averaging within groups first?

$$E(Y) = \frac{80 + 90 + 90 + 100 + 100 + 110 + 110 + 120}{8} = 100.$$

Invented population. Partition intuition: Ben Lambert, The Law of Iterated Expectations: an introduction.

Back to main lecture

LIE: worked example

Average within sex groups first — a larger group must get more weight

Male: 2 people

A	B
80	90

Female: 6 people

C	D	E	F	G	H
90	100	100	110	110	120

Invented population and group memberships; IQ scores are in points.

Back to main lecture

LIE: worked example

Average within sex groups first — a larger group must get more weight

Male: 2 people

A	B
80	90

Mean: 85

Female: 6 people

C	D	E	F	G	H
90	100	100	110	110	120

Mean: 105

Is the overall mean $(85 + 105)/2 = 95$**?**

Invented population and group memberships; IQ scores are in points.

Back to main lecture

LIE: worked example

Average within sex groups first — a larger group must get more weight

Male: 2 people

A	B
80	90

Mean: 85

Female: 6 people

C	D	E	F	G	H
90	100	100	110	110	120

Mean: 105

Is the overall mean $(85 + 105)/2 = 95$**?**

$$E(Y) = \frac{2}{8} \times 85 + \frac{6}{8} \times 105 = 100.$$

Equal weights overrepresent people in the smaller group.

Invented population and group memberships; IQ scores are in points.

Back to main lecture

LIE: worked example

Regroup by age — the overall mean stays 100

Under 40: 4 people

A	C	D	F
80	90	100	110

40 or older: 4 people

B	E	G	H
90	100	110	120

Invented population. Every person appears in exactly one age group, including age 40 in the second group.

Back to main lecture

LIE: worked example

Regroup by age — the overall mean stays 100

Under 40: 4 people

A	C	D	F
80	90	100	110

Mean: 95

40 or older: 4 people

B	E	G	H
90	100	110	120

Mean: 105

$$E(Y) = \frac{4}{8} \times 95 + \frac{4}{8} \times 105 = 100.$$

Invented population. Every person appears in exactly one age group, including age 40 in the second group.

Back to main lecture

LIE: worked example

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A	C	D	F
80	90	100	110

Mean: 95

40 or older: 4 people

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90	100	110	120

Mean: 105

$$E(Y) = \frac{4}{8} \times 95 + \frac{4}{8} \times 105 = 100.$$

Regrouping changes the means and weights, but preserves the overall mean.

Invented population. Every person appears in exactly one age group, including age 40 in the second group.

Back to main lecture

LIE: worked example

Unconditional LIE — read the inner and outer expectations

For a grouping variable W, let $\mu(w) = E(Y \mid W = w)$.

Discrete-group version, with finite $E(|Y|)$. For continuous W, the outer average is an integral.

Back to main lecture

LIE: worked example

Unconditional LIE — read the inner and outer expectations

For a grouping variable W, let $\mu(w) = E(Y \mid W = w)$.

$$E[E(Y \mid W)] = \sum_{w} \underbrace{P(W = w)}_{\text{group share}} \underbrace{\mu(w)}_{\text{group mean}} = E(Y).$$

Discrete-group version, with finite $E(|Y|)$. For continuous W, the outer average is an integral.

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LIE: worked example

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$$E[E(Y \mid W)] = \sum_{w} \underbrace{P(W = w)}_{\text{group share}} \underbrace{\mu(w)}_{\text{group mean}} = E(Y).$$

$$E[E(Y \mid \textit{sex})] = E[E(Y \mid \textit{age group})] = 100.$$

Use every group and its correct probability. Independence is not required.

Discrete-group version, with finite $E(|Y|)$. For continuous W, the outer average is an integral.

▸ Visual: balance point ▸ Sample: NLS80

◂ Back to main lecture

Properties of conditional expectations

Wooldridge App. 2A, CE.4 — pull a known factor outside the average

$$E[Yg(\mathbf{x}) \mid \mathbf{x}] = g(\mathbf{x})E(Y \mid \mathbf{x}).$$

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Within a conditioning group, everyone has the same value of $g(\mathbf{x})$.

Properties of conditional expectations

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$$E[Yg(\mathbf{x}) \mid \mathbf{x}] = g(\mathbf{x})E(Y \mid \mathbf{x}).$$

Within a conditioning group, everyone has the same value of $g(\mathbf{x})$.

$$E[educ \cdot Y \mid educ] = educ \cdot E(Y \mid educ).$$

This rule lets us manipulate interactions and conditional-mean errors.

Back to main lecture

Properties of conditional expectations

Wooldridge (2.27)–(2.29) — transformations do not add information

Suppose $E(y \mid \mathbf{x}) = g[f(\mathbf{x})]$ and define $\mathbf{z} = f(\mathbf{x})$.

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Properties of conditional expectations

Wooldridge (2.27)–(2.29) — transformations do not add information

Suppose $E(y \mid \mathbf{x}) = g[f(\mathbf{x})]$ and define $\mathbf{z} = f(\mathbf{x})$.

$$E(y \mid \mathbf{z}) = E(y \mid \mathbf{x}) = g(\mathbf{z}).$$

- Adding e^2 and se to (s, e) does not change what is known.
- More generally, $f(\mathbf{x})$ may compress information. Replacing $\mathbf{x}$ by $f(\mathbf{x})$ preserves the CE here because the assumed mean depends only on that function.

Back to main lecture

Conditional (nested) LIE: wages

Hold schooling and experience fixed — average over unobserved ability

$$E(Y \mid \mathbf{x}) = E\big[E(Y \mid \mathbf{x}, q) \mid \mathbf{x}\big] .$$

Conditional (nested) LIE: wages

Hold schooling and experience fixed — average over unobserved ability

$$E(Y \mid \mathbf{x}) = E\big[E(Y \mid \mathbf{x}, q) \mid \mathbf{x}\big].$$

- Inner average: wages within schooling, experience *and ability* groups.
- Outer average: average those means over ability *within each schooling–experience group*.

Conditional (nested) LIE: wages

Hold schooling and experience fixed — average over unobserved ability

$$E(Y \mid \mathbf{x}) = E\big[E(Y \mid \mathbf{x}, q) \mid \mathbf{x}\big].$$

- Inner average: wages within schooling, experience *and ability* groups.
- Outer average: average those means over ability *within each schooling–experience group*.
- This links a CE involving an unobserved q to a CE involving only observed $\mathbf{x}$.

The outer weights are the distribution of q given $\mathbf{x}$, not necessarily its overall population distribution.

▸ Rule: which information set survives?

◂ Back to main lecture

LIE: sample analogue

NLS80, 935 men — the same weighting rule works in a sample

Schooling group	Men	Mean wage (US$)
9–11 years	88	774.25
12 years	393	862.67
13–18 years	454	1,076.02

Can we give these means equal weight?

Sample illustration of LIE: weight each group mean by its share. `nls80.dta`; rounded means, exact totals.

Back to main lecture

LIE: sample analogue

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Equal weights give \$904.32, but the groups have different sizes.

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Equal weights give \$904.32, but the groups have different sizes.

$$\bar{Y} = \sum_g \frac{n_g}{935} \bar{Y}_g = \frac{68{,}134 + 339{,}030 + 488{,}515}{935} = 957.95.$$

Sample illustration of LIE: weight each group mean by its share. `nls80.dta`; rounded means, exact totals.

Back to main lecture

LIE: which information set survives?

Wooldridge p. 20 — "the smaller information set always dominates"

Y: wage; X: sex; Z: age group. Conditioning on a variable means knowing it.

Iterated expectation	Equals	Averaged out
$E[\,E(Y \mid X, Z) \mid Z\,]$	$E(Y \mid Z)$	sex, within each age group
$E[\,E(Y \mid X, Z) \mid X\,]$	$E(Y \mid X)$	age, within each sex
$E[\,E(Y \mid X, Z)\,]$	$E(Y)$	both
$E[\,E(Y \mid Z) \mid X, Z\,]$	$E(Y \mid Z)$	nothing: already known given (X, Z)

Wooldridge (2.19)–(2.21); App. 2A, CE.3. Row four is (2.20): the inner mean is already known given (X, Z).

Back to main lecture

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$E[\,E(Y \mid Z) \mid X, Z\,]$	$E(Y \mid Z)$	nothing: already known given (X, Z)

Rule: for nested information sets, the less informative set survives. The dropped variable is averaged with its distribution *given* the kept ones.

Wooldridge (2.19)–(2.21); App. 2A, CE.3. Row four is (2.20): the inner mean is already known given (X, Z).

Back to main lecture

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$E[\,E(Y \mid X, Z)\,]$	$E(Y)$	both
$E[\,E(Y \mid Z) \mid X, Z\,]$	$E(Y \mid Z)$	nothing: already known given (X, Z)
$E[\,E(Y \mid X) \mid Z\,]$	no simplification	information sets need not be nested

Rule: for nested information sets, the less informative set survives. The dropped variable is averaged with its distribution *given* the kept ones.

Wooldridge (2.19)–(2.21); App. 2A, CE.3. Row four is (2.20): the inner mean is already known given (X, Z).

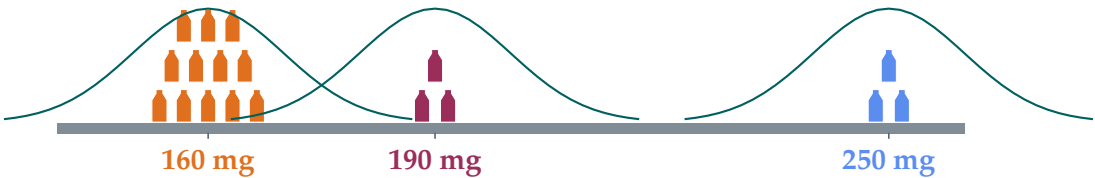
Back to main lecture

APPENDIX D · Wooldridge App. 2A

Variance decomposition: the bottle story continued

Variance decomposition: the bottle story

Every city has SD 10 mg — what is the overall SD of a random bottle?



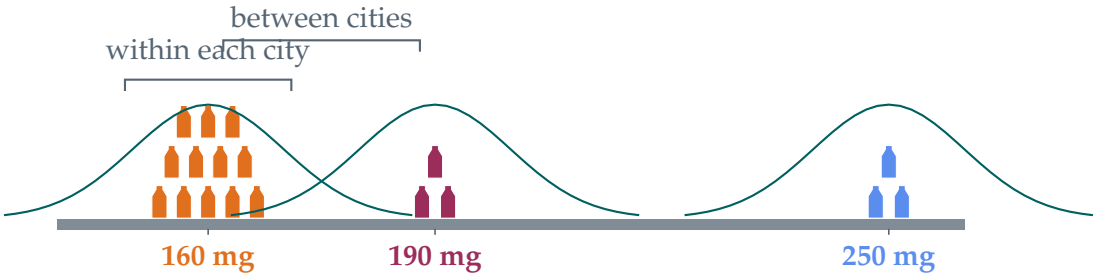
Each city: SD 10 mg around its mean. The overall SD of a random bottle is
A) 10 mg B) more C) less D) it depends

Same bottle shares; curves illustrate within-city spread. Adapted from HarvardX, Stat 110.

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Variance decomposition: the bottle story

Every city has SD 10 mg — what is the overall SD of a random bottle?



Bottles vary *within* each city; city means vary *between* cities. Is total variance just the weighted average of the city variances?

Same bottle shares; curves illustrate within-city spread. Adapted from HarvardX, Stat 110.

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Variance decomposition: the bottle story

Average of the separate variances, plus variance of the separate averages

$$\text{Var}(Y) = \underbrace{E[\text{Var}(Y \mid X)]}_{\text{average of the separate variances}} + \underbrace{\text{Var}[E(Y \mid X)]}_{\text{variance of the separate averages}}$$

Adapted bottle shares $2/3, 1/6, 1/6$ throughout. Wooldridge App. 2A, CV.2.

Back to main lecture

Variance decomposition: the bottle story

Average of the separate variances, plus variance of the separate averages

$$\text{Var}(Y) = \underbrace{E[\text{Var}(Y \mid X)]}_{\text{average of the separate variances}} + \underbrace{\text{Var}[E(Y \mid X)]}_{\text{variance of the separate averages}}$$

Within: every city has variance $10^2 = 100$, so the weighted average is

$$\tfrac{2}{3}(100) + \tfrac{1}{6}(100) + \tfrac{1}{6}(100) = 100 \text{ mg}^2.$$

Adapted bottle shares 2/3, 1/6, 1/6 throughout. Wooldridge App. 2A, CV.2.

Back to main lecture

Variance decomposition: the bottle story

Average of the separate variances, plus variance of the separate averages

$$\text{Var}(Y) = \underbrace{E[\text{Var}(Y \mid X)]}_{\text{average of the separate variances}} + \underbrace{\text{Var}[E(Y \mid X)]}_{\text{variance of the separate averages}}$$

Between: the city means $160, 190, 250$ around the overall mean 180, with the bottle shares:

$$\tfrac{2}{3}(160-180)^2 + \tfrac{1}{6}(190-180)^2 + \tfrac{1}{6}(250-180)^2 = 1100 \text{ mg}^2.$$

Adapted bottle shares $2/3, 1/6, 1/6$ throughout. Wooldridge App. 2A, CV.2.

Back to main lecture

Variance decomposition: the bottle story

Average of the separate variances, plus variance of the separate averages

$$\text{Var}(Y) = \underbrace{E[\text{Var}(Y \mid X)]}_{\text{average of the separate variances}} + \underbrace{\text{Var}[E(Y \mid X)]}_{\text{variance of the separate averages}}$$

Total: $\text{Var}(Y) = 100 + 1100 = 1200$ mg^2, so

$$SD(Y) = \sqrt{1200} \approx 35 \text{ mg} > 10 \text{ mg}.$$

With multiple sources of water, there are multiple sources of variation.

Adapted bottle shares 2/3, 1/6, 1/6 throughout. Wooldridge App. 2A, CV.2.

Back to main lecture

Law of total variance

Wooldridge App. 2A, CV.2 — total = within + between (Eve's law)

Y: outcome; X: group; $\mu(X) = E(Y \mid X)$.

Within groups: outcomes differ from their own group mean.

$$E[\mathrm{Var}(Y \mid X)] = E\Big[(Y - \mu(X))^2\Big].$$

Wooldridge, App. 2A, CV.2. Finite second moment. Use group probabilities; standard deviations do not add.

Back to main lecture

Law of total variance

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$$\mathrm{Var}[E(Y \mid X)] = E\Big[(\mu(X) - E(Y))^2\Big].$$

Wooldridge, App. 2A, CV.2. Finite second moment. Use group probabilities; standard deviations do not add.

Back to main lecture

Law of total variance

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$$\boxed{\mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid X)] + \mathrm{Var}[E(Y \mid X)]}.$$

Wooldridge, App. 2A, CV.2. Finite second moment. Use group probabilities; standard deviations do not add.

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APPENDIX E · Wooldridge 2.2.4–2.2.5

Omitted variables and proxies

Unobserved regressor: the error-form route

Why LIE is needed — start from the error form instead of the conditional mean

ERROR FORM; ABILITY q IS UNOBSERVED; TARGET SLOPES IN BLUE

$$y = \beta_0 + \beta_s s + \beta_e e + \gamma q + v, \qquad E(v \mid s, e, q) = 0.$$

Take $E(\cdot \mid s, e)$ of both sides; linearity leaves conditional expectations of ability and the error:

Unobserved regressor: the error-form route

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Take $E(\cdot \mid s, e)$ of both sides; linearity leaves conditional expectations of ability and the error:

$$E(y \mid s, e) = \beta_0 + \beta_s s + \beta_e e + \gamma\, E(q \mid s, e) + E(v \mid s, e).$$

Only $E(v \mid s, e, q) = 0$ was assumed. Why is $E(v \mid s, e)$ zero as well?

Back to main lecture

Unobserved regressor: the error-form route

The error's mean on the coarser information: one application of LIE

ASSUMED ON THE FINER INFORMATION (s, e, q)

$$E(v \mid s, e, q) = 0.$$

Unobserved regressor: the error-form route

The error's mean on the coarser information: one application of LIE

ASSUMED ON THE FINER INFORMATION (s, e, q)

$$E(v \mid s, e, q) = 0.$$

$$E(v \mid s, e) = E[\, E(v \mid s, e, q) \mid s, e \,] = E[\, 0 \mid s, e \,] = 0.$$

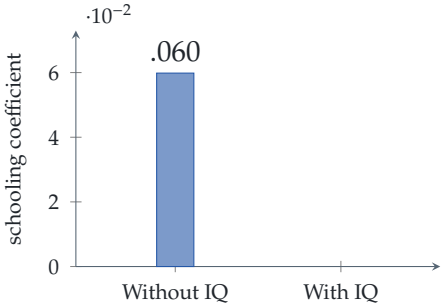
Zero conditional mean on the finer information (s, e, q) implies zero on the coarser (s, e). The extra term drops out, and the main-route equation results.

Wooldridge 2.2.3, (2.15)–(2.16) define the error; 2.2.4, (2.21) is the LIE used; App. 2A, CE.3.

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Unobserved regressor: real data

NLS80 — why adding IQ changes the schooling coefficient



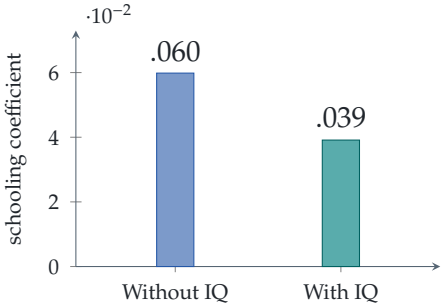
- Outcome: log(*wage*). Regressors: schooling, with or without IQ.

Wooldridge Example 4.3 data; `nls80.dta`; regressions include an intercept.

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Unobserved regressor: real data

NLS80 — why adding IQ changes the schooling coefficient



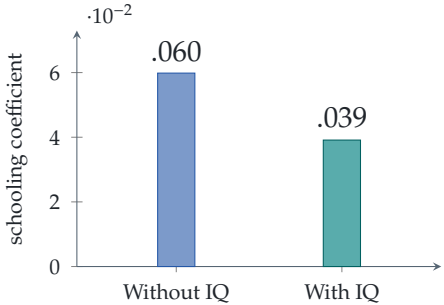
- Outcome: log(*wage*). Regressors: schooling, with or without IQ.
- Adding IQ changes the schooling coefficient from .060 to .039.

Wooldridge Example 4.3 data; `nls80.dta`; regressions include an intercept.

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Unobserved regressor: real data

NLS80 — why adding IQ changes the schooling coefficient



- Outcome: log(*wage*). Regressors: schooling, with or without IQ.
- Adding IQ changes the schooling coefficient from .060 to .039.
- Identity: $.03912 + .005863 \times 3.53383 = .05984$.

Omitted-variable algebra at work; no proof that IQ is a valid proxy.

Wooldridge Example 4.3 data; `nls80.dta`; regressions include an intercept.

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Unobserved regressor: interactions

Wooldridge 2.2.4, (2.26) — omitted interactions can create curvature

SCHOOLING s, EXPERIENCE e, UNOBSERVED ABILITY q

$$E(y \mid s, e, q) = \beta_0 + \beta_s s + \beta_e e + \gamma q + \theta s q,$$
$$E(q \mid s, e) = \delta_0 + \delta_s s + \delta_e e.$$

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$$E(q \mid s, e) = \delta_0 + \delta_s s + \delta_e e.$$

$$E(y \mid s, e) = \beta_0 + \beta_s s + \beta_e e + (\gamma + \theta s)(\delta_0 + \delta_s s + \delta_e e).$$

Unobserved regressor: interactions

Wooldridge 2.2.4, (2.26) — omitted interactions can create curvature

SCHOOLING s, EXPERIENCE e, UNOBSERVED ABILITY q

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$$E(q \mid s, e) = \delta_0 + \delta_s s + \delta_e e.$$

$$E(y \mid s, e) = \beta_0 + \beta_s s + \beta_e e + (\gamma + \theta s)(\delta_0 + \delta_s s + \delta_e e).$$

Expanding introduces s^2 and se. More flexible fitting can describe the observed CE without recovering the structural schooling effect.

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Proxy variables with interactions

Reading ahead — Lecture 3 and Wooldridge Chapter 4

If ability interacts with schooling, the structural partial effect varies with ability:

$$E(y \mid s, e, q) = \beta_0 + \beta_s s + \beta_e e + \gamma q + \theta s q, \qquad PE_s = \beta_s + \theta q.$$

Proxy variables with interactions

Reading ahead — Lecture 3 and Wooldridge Chapter 4

If ability interacts with schooling, the structural partial effect varies with ability:

$$E(y \mid s, e, q) = \beta_0 + \beta_s s + \beta_e e + \gamma q + \theta s q, \qquad PE_s = \beta_s + \theta q.$$

The structural APE is $\beta_s + \theta E(q)$; recovering it from a proxy requires more algebra than the additive example.

- Read Wooldridge Chapter 4 on proxy variables.
- Lecture 3 develops the schooling–ability interaction and proxy calculation.

APPENDIX F · Wooldridge 2.3

Linear projections

Linear projections

Wooldridge 2.3, (2.38)–(2.42) — the best linear predictor of a curved mean

Which line, when conditional means are curved? The linear projection:

$$L(y \mid 1, \mathbf{x}) = b_0 + \mathbf{x}b.$$

Linear projections

Wooldridge 2.3, (2.38)–(2.42) — the best linear predictor of a curved mean

Which line, when conditional means are curved? The linear projection:

$$L(y \mid 1, \mathbf{x}) = b_0 + \mathbf{x}b.$$

Its coefficients minimise $E[(y - b_0 - \mathbf{x}b)^2]$, the mean squared prediction error. Best among linear functions of the regressors; it need not equal the CE.

Linear projections

Wooldridge 2.3, (2.38)–(2.42) — the best linear predictor of a curved mean

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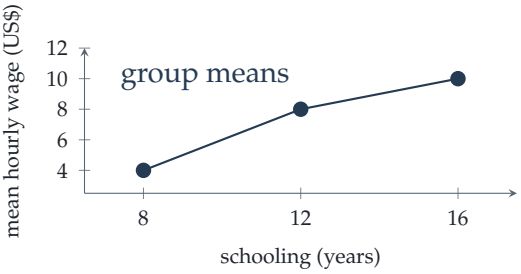
$$b_1 = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)}, \qquad b_0 = E(y) - b_1 E(x) \qquad \text{(one regressor)}.$$

Finite second moments, nonsingular $\mathrm{Var}(\mathbf{x})$; general case $b = [\mathrm{Var}(\mathbf{x})]^{-1}\mathrm{Cov}(\mathbf{x}, y)$; $v = y - L(y \mid 1, \mathbf{x})$ has $E(v) = 0$ and $\mathrm{Cov}(x_j, v) = 0$.

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Linear projections: worked example

The best line can make systematic group errors



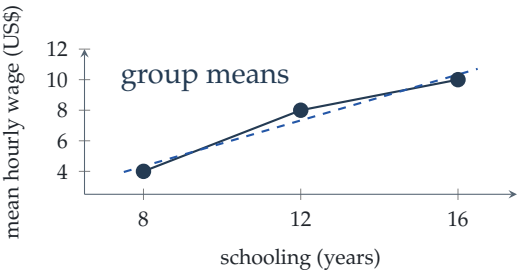
- Each schooling group has probability 1/3; CE means are 4, 8, 10.

Illustrative population. Connecting group means is a visual guide.

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Linear projections: worked example

The best line can make systematic group errors



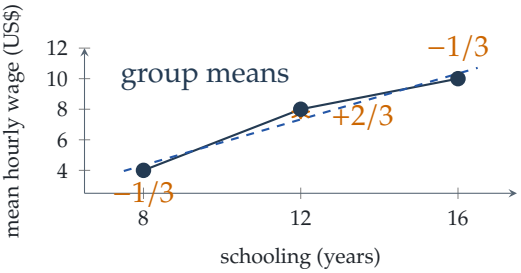
- Each schooling group has probability 1/3; CE means are $4, 8, 10$.
- Best line: $L(Y \mid 1, X) = -5/3 + .75X$.

Illustrative population. Connecting group means is a visual guide.

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Linear projections: worked example

The best line can make systematic group errors



- Each schooling group has probability 1/3; CE means are $4, 8, 10$.
- Best line: $L(Y \mid 1, X) = -5/3 + .75X$.
- CE matches each mean. The best line misses them.

With $v = Y - L(Y \mid 1, X)$: $E(v) = 0$, but $E(v \mid X = 12) = 2/3$.

Illustrative population. Connecting group means is a visual guide.

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